

Random Samples and Probability Models

Probability models for one event, estimates from a random sample, and summaries of the sample itself. Every data set below comes from a stated sample, and every count and unit is given in the problem.

- For equally likely outcomes, $P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{number of possible outcomes}}$. Leave probabilities as fractions in lowest terms.
- A random sample stands in for its population: the fraction in the sample is your estimate of the fraction in the whole population.
- Show the computation behind each answer, not just the result.

1. A raffle drum holds 20 tickets, all the same size and shape, and 8 of them are blue. One ticket is drawn without looking. Find the probability that the ticket drawn is blue. Give your answer as a fraction in lowest terms.

Answer: _____

2. A teacher picked a random sample of 8 students from her class roster and recorded how many minutes each spent on homework last night.

15	20	25	30	20	35	20	15
----	----	----	----	----	----	----	----

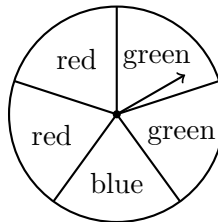
(a) Find the mean number of minutes for this sample. (b) Find the mode.

Answer: _____

3. To estimate how many fish live in a pond, a biologist tags 12 fish and releases them. A week later she nets a random sample of 40 fish and finds that 6 of them are tagged. **(a)** What fraction of the sample was tagged? Give it in lowest terms. **(b)** Assuming the same fraction of the whole pond is tagged, estimate how many fish live in the pond.

Answer: _____

4. A prize spinner has five equal sections, shown below. It is spun once. **(a)** Find the probability that it does *not* land on red. **(b)** Find the probability that it lands on green.



Answer: _____

5. A town surveyed a random sample of 12 households and recorded the number of pets in each.

0	0	1	1	1	2	2	2	3	3	4	5
---	---	---	---	---	---	---	---	---	---	---	---

(a) Find the median number of pets per household in this sample. **(b)** Find the total number of pets in the sampled households.

Answer: _____

6. A middle school has 900 students. The principal draws a random sample of 50 students from the enrolment list and finds that 14 of them walk to school. **(a)** What fraction of the sample walks to school, in lowest terms? **(b)** Use that fraction to estimate how many students in the whole school walk to school.

Answer: _____

7. A fair number cube has faces numbered one through six. **(a)** Find the probability of rolling a four on one roll. **(b)** The cube will be rolled 300 times. Use the probability model to predict how many of those rolls land on four.

Answer: _____

8. A weather club recorded the daily high temperature, in degrees Fahrenheit, on seven randomly chosen days in April.

12	15	15	18	22	27	31
----	----	----	----	----	----	----

(a) Find the median of the sample. **(b)** Find the interquartile range — the spread of the middle half of the sample. A classmate answers 19 for part (b); explain what that number is instead.

Answer: _____

9. A student wants to know what fraction of her school's 800 students prefer skateboarding to biking. She asks 60 students at the skate park after school, and 45 of them say skateboarding. **(a)** What fraction of her sample prefers skateboarding, in lowest terms? **(b)** What estimate for the whole school does that fraction give? **(c)** Explain why this estimate should not be trusted, and describe a way of choosing the sample that would make it trustworthy.

Answer: _____

10. A cereal company puts one of four equally likely prize cards in every box, and you want to know the chance of getting the card you want in your next box. Design a simulation that models this situation. Say what random device you would use, which outcomes stand for getting the card you want, what counts as one trial, how many trials you would run, and how you would use the results to estimate the probability.

Answer: _____